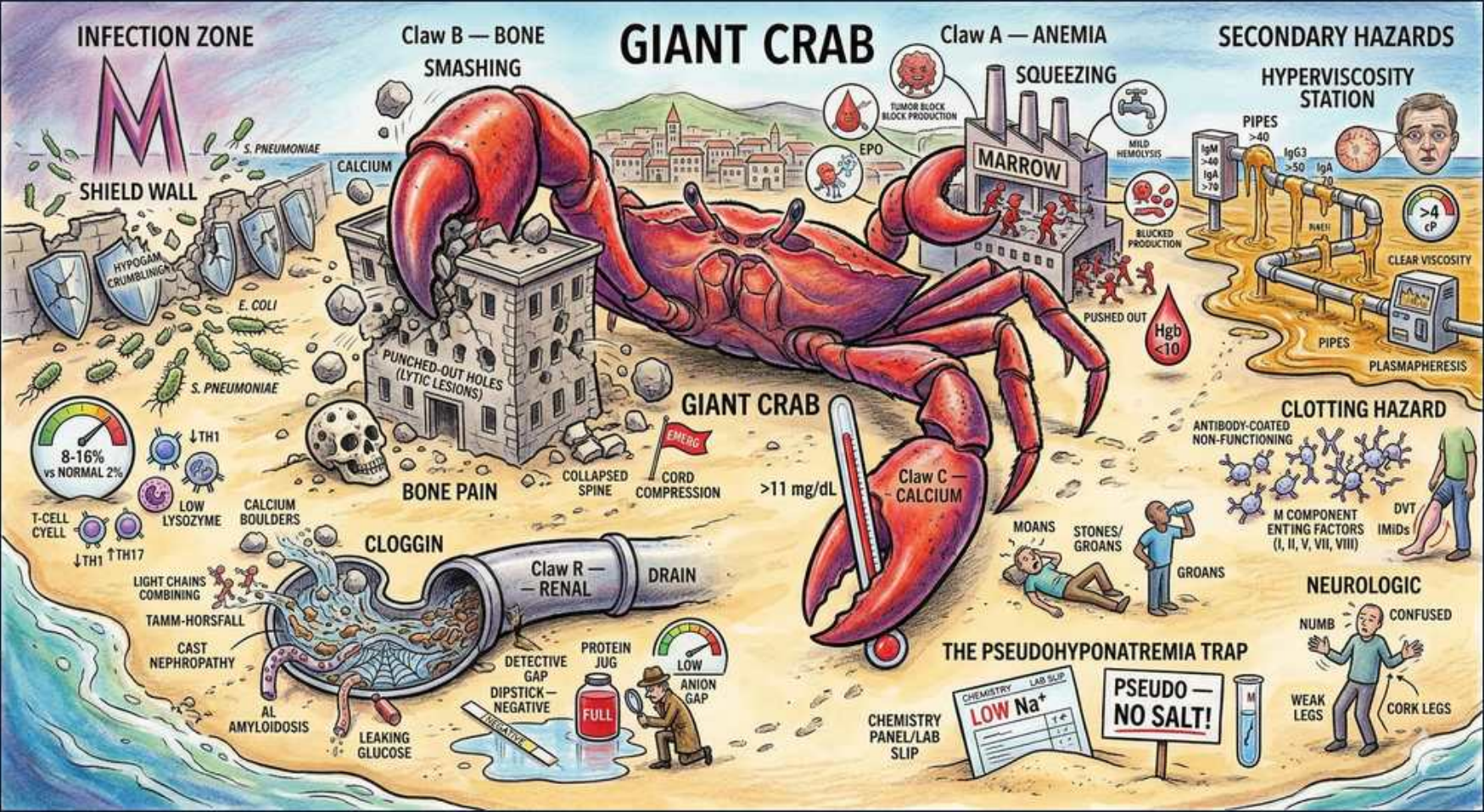


Plasma Cell — Giant Crab (CRAB Criteria & Complications)



1. M = monoclonal protein

WHAT YOU SEE

A large purple letter M standing at the top-left of the scene beside the SHIELD WALL — the M is the monoclonal protein, the engine whose clone drives every CRAB consequence around it.

WHAT IT ENCODES

The monoclonal protein from clonal plasma cells is the underlying driver; CRAB are its end-organ consequences (hyperCalcemia, Renal failure, Anemia, Bone lesions). Active multiple myeloma = clonal marrow plasma cells (usually ≥10%) PLUS ≥1 myeloma-defining event (CRAB or a SLiM biomarker). Mnemonic: the M-protein is the engine; CRAB is the wreckage.

BOARD TRAP

WRONG answer: diagnosing myeloma from the M-protein alone without a defining event. The discriminator is that an M-protein with no CRAB and no SLiM biomarker is MGUS or smoldering myeloma — you need a CRAB feature or a SLiM criterion (60% plasma cells, FLC ratio ≥100, >1 MRI lesion) to call it active myeloma.

2. Claw C — Calcium greater than 11

WHAT YOU SEE

Claw C CALCIUM, greater than 11 mg/dL flag

WHAT IT ENCODES

CRAB 'C' = hyperCalcemia: serum calcium >11 mg/dL (or >1 mg/dL above the upper normal), driven by osteoclast-activating factors (RANKL) causing bone resorption. Symptoms are the classic 'stones, bones, groans, psychiatric overtones.' Treat with aggressive IV normal saline first, then a bisphosphonate (zoledronic acid) and calcitonin for rapid effect.

BOARD TRAP

WRONG answer: managing myeloma hypercalcemia with loop diuretics (or thiazides) up front. The discriminator is volume status — give IV saline FIRST; thiazides RAISE calcium, and loops are only added after the patient is volume-replete (not as primary therapy).

3. Claw R — Renal

WHAT YOU SEE

Claw R RENAL with DRAIN/CLOGGIN pipe

WHAT IT ENCODES

CRAB 'R' = Renal insufficiency: creatinine >2 mg/dL or creatinine clearance <40 mL/min, most often from light-chain cast nephropathy ('myeloma kidney'). Hypercalcemia, dehydration, hyperuricemia, NSAIDs, and IV contrast all worsen it. Management: hydration, treat hypercalcemia, rapid anti-myeloma therapy (bortezomib-based) to lower light chains.

BOARD TRAP

WRONG answer: ordering a contrast CT/IVP or giving NSAIDs to evaluate/treat the renal failure. The discriminator is light-chain cast nephropathy — IV contrast and NSAIDs precipitate casts and worsen AKI; avoid them and instead hydrate and reduce light-chain production.

4. Claw A — Anemia

WHAT YOU SEE

Claw A ANEMIA, marrow factory, EPO

WHAT IT ENCODES

CRAB 'A' = Anemia: hemoglobin <10 g/dL or >2 g/dL below normal, typically normocytic-normochromic from marrow replacement by plasma cells plus relative erythropoietin deficiency (worse with renal failure). Rouleaux on smear and a high ESR are classic clues. Address with treating myeloma; EPO/transfusion as supportive care.

BOARD TRAP

WRONG answer: working it up as iron-deficiency anemia and giving iron. The discriminator is a normocytic anemia with rouleaux, high ESR/total protein, and an M-spike — this is anemia of marrow infiltration/chronic disease, not iron deficiency; pursue SPEP/marrow rather than iron studies alone.

5. Claw B — Bone smashing

WHAT YOU SEE

Claw B BONE SMASHING crab claw crushing buildings

WHAT IT ENCODES

CRAB 'B' = Bone disease: lytic ('punched-out') lesions, osteopenia, or pathologic fractures from RANKL-driven osteoclast activation with suppressed osteoblasts. Best detected by whole-body low-dose CT, PET-CT, or MRI — NOT bone scan. Bisphosphonates (zoledronic acid) or denosumab reduce skeletal events.

BOARD TRAP

WRONG answer: ordering a technetium-99m bone scan to find myeloma lesions. The discriminator is that myeloma is purely osteolytic with little osteoblastic activity, so bone scans are falsely negative — use skeletal survey/whole-body CT, PET-CT, or MRI instead.

6. Punched-out lytic lesions

WHAT YOU SEE

COLLAPSED SPINE, punched-out holes, CORD COMPRESSION flag

WHAT IT ENCODES

Lytic lesions cause bone pain, pathologic fractures, and vertebral compression that can produce spinal cord compression — an oncologic emergency presenting with back pain, leg weakness, sensory level, and bladder/bowel dysfunction. Immediate management: emergent MRI plus high-dose IV dexamethasone, then radiation or surgical decompression. Mnemonic: new back pain + weak legs in myeloma = image the cord now.

BOARD TRAP

WRONG answer: attributing acute back pain with leg weakness to musculoskeletal strain and ordering plain films/physical therapy. The discriminator is cord compression — it requires urgent MRI of the whole spine and steroids/radiation; delay risks permanent paralysis.

7. Cast nephropathy / myeloma kidney

WHAT YOU SEE

CAST NEPHROPATHY, TAMM-HORSFALL, light chains in tubule

WHAT IT ENCODES

Filtered monoclonal free light chains bind Tamm-Horsfall (uromodulin) protein in the distal tubule, forming obstructing waxy casts = cast nephropathy, the most common cause of myeloma renal failure. Reducing light-chain load fast (bortezomib-based therapy) and hydration are key. Mnemonic: light chains + Tamm-Horsfall = tubular traffic jam.

BOARD TRAP

WRONG answer: assuming the proteinuria is glomerular/albumin and that the urine dipstick reflects it. The discriminator is that cast nephropathy is tubulointerstitial from light chains (dipstick-negative) — quantify with UPEP/FLC; predominant ALBUMINURIA instead points to AL amyloid or light-chain deposition disease, not cast nephropathy.

8. Dipstick negative for light chains

WHAT YOU SEE

DIPSTICK NEGATIVE detective, PROTEIN LOW, LEAKING GLUCOSE

WHAT IT ENCODES

The urine dipstick detects ALBUMIN only, so monoclonal free light chains (Bence-Jones protein) read NEGATIVE despite heavy light-chain excretion — a classic discordance with a high 24-hour protein. Detect light chains with UPEP/immunofixation or serum FLC. (Proximal tubular injury can add Fanconi-type glucosuria with normal glucose.) Mnemonic: dipstick is blind to light chains.

BOARD TRAP

WRONG answer: excluding significant proteinuria/myeloma kidney because the dipstick is negative or trace. The discriminator is that dipstick misses light chains — order a 24-hour UPEP with immunofixation (or serum FLC) when you suspect myeloma.

9. AL amyloidosis

WHAT YOU SEE

AL AMYLOIDOSIS label near light chain cells

WHAT IT ENCODES

Monoclonal light chains (usually lambda) can misfold and deposit as AL amyloid, causing nephrotic-range ALBUMINURIA, restrictive/infiltrative cardiomyopathy (low-voltage ECG with thick walls on echo), peripheral/autonomic neuropathy, macroglossia, and easy bruising. Diagnose with a Congo-red biopsy (apple-green birefringence) plus typing. Mnemonic: AL = Amyloid Light-chain.

BOARD TRAP

WRONG answer: confusing AL amyloid kidney (heavy albuminuria, nephrotic syndrome) with cast nephropathy (light-chain casts, minimal albumin). The discriminator is the proteinuria type and organ pattern — albuminuria/nephrotic + restrictive cardiomyopathy + macroglossia means AL amyloidosis (confirm with Congo-red apple-green birefringence).

10. Infection susceptibility

WHAT YOU SEE

SHIELD WALL with *S. pneumoniae*, *E. coli*, encapsulated organisms

WHAT IT ENCODES

Despite a high total Ig, functional hypogammaglobulinemia (the malignant clone suppresses normal polyclonal antibody) plus neutropenia/treatment effects leaves patients vulnerable to encapsulated organisms (*S. pneumoniae*, *H. influenzae*) and gram-negatives like *E. coli* — infection is a leading cause of death in myeloma. Vaccinate (pneumococcal, influenza) and consider prophylaxis. Mnemonic: many antibodies, none that work.

BOARD TRAP

WRONG answer: assuming a high total immunoglobulin/M-protein means the patient is immunologically protected. The discriminator is that the M-protein is non-functional and normal Ig is suppressed — these patients are functionally immunodeficient and need infection prophylaxis and vaccination.

11. Hyperviscosity

WHAT YOU SEE

HYPERVISCOSITY STATION clogged pipes, plasmapheresis

WHAT IT ENCODES

Hyperviscosity from a high paraprotein causes headache, blurred/visual changes, mucosal bleeding, and confusion; in myeloma it is most associated with IgA (which polymerizes) or very high IgG. Emergency treatment is plasmapheresis to remove the paraprotein, then systemic therapy. Mnemonic: bleeding + blurring + blundering = check serum viscosity.

BOARD TRAP

WRONG answer: expecting hyperviscosity to be a common, typical feature of ordinary IgG myeloma. The discriminator is that hyperviscosity is classically Waldenström macroglobulinemia (IgM pentamer); in myeloma it is uncommon and points to IgA or markedly elevated IgG — treat with plasmapheresis regardless of subtype.

12. Clotting / coagulation hazard

WHAT YOU SEE

CLOTTING HAZARD, M component coating clotting factors

WHAT IT ENCODES

The paraprotein can coat platelets and interfere with fibrin/coagulation factors (acquired bleeding tendency), while the disease itself plus immunomodulatory drugs (IMiDs — thalidomide, lenalidomide, pomalidomide) markedly raise VTE risk. IMiD regimens require thromboprophylaxis (aspirin for low risk; LMWH/DOAC for higher risk). Mnemonic: myeloma can both bleed and clot.

BOARD TRAP

WRONG answer: starting lenalidomide-based therapy without any VTE prophylaxis. The discriminator is the high thrombosis risk of IMiDs — all patients on IMiD regimens need risk-adapted prophylaxis (aspirin vs anticoagulation), not just routine monitoring.

13. Neurologic compression

WHAT YOU SEE

NEUROLOGIC, NUMB, CONFUSED, WEAK LEGS

WHAT IT ENCODES

Neurologic problems in myeloma include spinal cord/nerve root compression from vertebral lesions (weak legs, sensory level — an emergency), confusion from hypercalcemia or hyperviscosity, and peripheral neuropathy from AL amyloid or bortezomib toxicity. New focal deficits demand urgent spinal MRI. Mnemonic: numb, confused, weak legs = image the spine and check calcium/viscosity.

BOARD TRAP

WRONG answer: attributing new leg weakness purely to chemotherapy-induced peripheral neuropathy. The discriminator is a myelopathic pattern (bilateral leg weakness, sensory level, bladder dysfunction) signaling CORD COMPRESSION — that needs emergent MRI, steroids, and radiation/surgery, not just dose-reducing bortezomib.

14. Pseudohyponatremia trap

WHAT YOU SEE

PSEUDOHYPONATREMIA TRAP, LOW Na, PSEUDO NO SALT

WHAT IT ENCODES

A very high paraprotein increases the non-aqueous (solid) fraction of plasma, so flame-photometry/indirect-ISE sodium reads falsely LOW — pseudohyponatremia — while measured serum OSMOLALITY is normal. There is no true sodium deficit and no symptoms; confirm with direct ion-selective electrode and serum osmolality. Mnemonic: 'pseudo, no salt' — the lab is fooled, the patient is fine.

BOARD TRAP

WRONG answer: treating the low sodium with hypertonic saline or fluid restriction. The discriminator is a NORMAL measured serum osmolality with a normal osmolar gap and high paraprotein — it is an artifact (pseudohyponatremia), so correct the measurement method, not the patient.
